

Long-term Treatment with Glucocorticoids/ACTH in Asthmatic Children

III. Effects on Growth and Adult Height

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ABSTRACT. Oberger, E., Engström, I. and Karlberg, J. (Department of Paediatrics, Karolinska Institute, Stockholm and Departments of Pediatrics I and Anatomy, Gothenburg University, Göteborg, Sweden). Longterm treatment with glucocorticoids/ACTH in asthmatic children. III. Effects on growth and adult height. *Acta Paediatr Scand* 79: 77, 1990.

The effect on growth of long-term treatment with prednisolone and/or ACTH_{1–24} (tetracosactrin) depot preparation was studied in 40 children with severe bronchial asthma. Height velocity was subnormal before treatment. During treatment the group of 17 children primarily treated with ACTH showed a moderate increase in mean velocity. Their height was not significantly altered, and neither was the age at peak height velocity nor adult height. In the group of 23 children treated with prednisolone the mean velocity decreased, resulting in a relative decrease in height. Peak height velocity was delayed by about 2 years in the boys but occurred at the expected time in the girls, as did menarche. Mean adult height was lower than expected after adjustment for mid-parenteral height. In 10 children ACTH was substituted for prednisolone, and their height velocity increased but not enough to affect adult height, which was just as low as in the patients treated with prednisolone only. *Key words:* bronchial asthma, children, adolescents, growth, peak height velocity, adult height, glucocorticoids, corticotrophin.

Inhibition of growth is a well-known side effect of long term systemic glucocorticoid treatment and occurs in children with bronchial asthma whether receiving alternate day (1, 2) or daily administration (3–7). After withdrawal of such treatment, growth accelerates (3, 6, 7), in particular during therapy with ACTH (8, 9). There are no reports on effects of long term treatment with glucocorticoids and/or ACTH on adult height. The aim of the present study was to investigate the immediate and the long term effect of glucocorticoids and/or ACTH on growth and timing of pubertal maturation. Longitudinal growth data were collected in children with severe bronchial asthma treated with prednisolone and/or ACTH_{1–24} (tetracosactrin) depot preparation and followed through the treatment period until the end of growth.

MATERIAL AND METHODS

Our series consist of 40 children, born in 1947–1974, with bronchial asthma severe enough to require long term treatment with glucocorticoids or ACTH (Table 1). Seventeen children were given ACTH without prior treatment with prednisolone (group A). Twenty-three children were given prednisolone (groups P and P+A). Ten of these were transferred to ACTH after a period of combined treatment (group P+A).

During the period of study ACTH was withdrawn in all children except one (in group P+A) and prednisolone was withdrawn in 8 of 13 children (in group P). All children in whom the treatment was withdrawn were followed further for ≥ 1 year, 32 until final height.

ACTH_{1–24} (Synacthen® Depot, CIBA) 0.2–0.5 mg (0.006–0.031 mg/kg/week), was given i.m. at intervals of 2–7 days. Prednisolone, 2.5 to 15 mg/day (0.08–0.56 mg/kg/day) was given daily in 2–3 divided doses. Each month, height was measured (10), the clinical status was assessed, and the

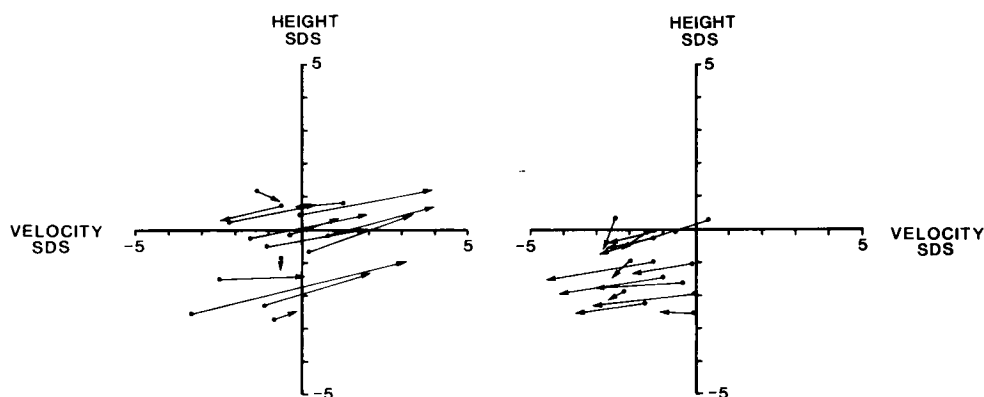


Fig. 1. Height SDS at the start of treatment in relation to height velocity SDS during the year before treatment (○) and height SDS after 1 year of treatment in relation to height velocity SDS during the first year of treatment (→) with ACTH (group A, left panel) and prednisolone (groups P+A and P, right panel).

dosage adjusted aiming at as low a dose as compatible with adequate control of asthma. For 30 of the 40 children height/data were available 1 year before treatment.

Calculations and statistical method. Height and annual height velocity were converted into standard deviation scores (SDS) (11). Peak height velocity (PHV) and menarche were used as maturity indices. PHV was individually estimated from annual height velocities. Age at PHV was equal to the visually observed maximum point of the curve of these velocities smoothed by a spline function (12) and was expressed in SDS. Adult height, defined as height after a height velocity of <0.5 cm/year, and adjusted for midparental height was expressed in SDS of the expected adult height as estimated from the linear regression fitted for data of adult height and midparental height for healthy Swedish children (publication forthcoming). All computations were performed at the mainframe computer center at the University of Gothenburg (13).

The study was approved by the Ethics Committee of the Karolinska Institute at Karolinska Hospital, Stockholm, Sweden.

Table 1. *Presentation of the patient series*

Group A: Children treated with ACTH^a only, Group P+A: Children treated with prednisolone and transferred to ACTH, Group P: Children treated with prednisolone only. Values are means (ranges) in years

Group . . .	A	P+A	P
Number of children . . .	17	10	13 ^b
Age at start of asthmatic symptoms	1.5 (0.5–4.0)	1.3 (0.3–2.0)	1.6 (0.3–3.0)
Age at start of treatment with prednisolone		4.8 (1.8–9.0)	7.3 (2.5–11.2)
Duration of treatment with prednisolone		7.0 (2.0–13.9)	6.4 (3.8–9.1)
			(n=8) ^c
Age at start of treatment with ACTH	5.4 (1.5–11.0)	10.6 (4.6–15.4)	
Duration of combined treatment with prednisolone and ACTH		1.2 (0.7–2.9)	
Duration of treatment with ACTH alone	3.1 (0.7–6.9)	3.7 (0.6–5.6)	
		(n=9)	

^a Synacthen® Depot.

^b 1/13 patients was included into the study during the 2nd year of treatment.

^c Prednisolone was withdrawn in 8/13 patients during the time of the study.

RESULTS

Growth. In the whole series ($n=40$) the mean height SDS was below the reference mean (11) ($p<0.001$) at start of treatment, and the mean height velocity SDS ($n=30$) was below the reference mean ($p<0.001$) during the year immediately preceding the treatment. Height velocity increased in all children but three during the first year of treatment with ACTH and decreased in all children during the first year with prednisolone (Fig. 1). Height velocity SDS remained at the same level during treatment with ACTH and decreased after its withdrawal ($p<0.01$) but the mean posttreatment value did not differ significantly from the reference mean (Table 2). No significant bivariate correlation was found for height velocity during the first year of treatment with the dose of ACTH, age, duration of the disease or height or height velocity before treatment.

During treatment with prednisolone, height velocity remained low, resulting in a progressive decrease in height SDS (Table 2). In the ten patients transferred from prednisolone to ACTH (group P+A) (Table 1) growth accelerated, but 1 year after withdrawal of prednisolone the mean height SDS still remained below the reference mean. Growth was retarded on doses of prednisolone as small as around 0.1 mg/kg/day (Fig. 2). No significant dose-response relationship was present. Furthermore, height velocity did not show any significant bivariate correlation with age, duration of the disease, or height or height velocity before treatment.

Puberty and adult height. Among the 17 children primarily treated with ACTH, 11 were followed through adolescence and 9 until adult height was reached. In this group neither

Table 2. Height SDS at start of treatment, at 1, 2 and 3 years of treatment and at 1 year after treatment withdrawal. Height velocity SDS during the year immediately preceding the treatment, years 1–3 of treatment and the year immediately following treatment withdrawal

Group A: Children treated with ACTH^a only, Group P+A: Children treated with prednisolone and transferred to ACTH, Group P: Children treated with prednisolone only. Values are mean $\pm$ SD; for the doses mean and range

	Group A			Groups P+A and P			Dose of prednisolone (mg/kg/day)	Dose of ACTH ^a (mg/kg/week)
	<i>n</i>	Height	Height velocity	<i>n</i>	Height	Height velocity		
Year before treatment	16	−0.52 ± 1.18	−0.96** ± 1.20	14	−1.04 ± 0.97	−0.98** ± 0.86		
1st year of treatment	17	−0.07 ± 0.99	1.09* ± 1.79	22	−1.39*** ± 0.70	−2.97*** ± 0.97	0.21 0.08–0.56	0.018 0.006–0.031
2nd year of treatment	12	0.24 ± 0.83	1.01* ± 1.52	21	−1.85*** ± 0.71	−2.39*** ± 1.40	0.20 0.07–0.54	0.014 0.006–0.023
3rd year of treatment	7	0.20 ± 1.01	1.01 ± 1.12	17	2.35*** ± 0.59	−1.94*** ± 1.44	0.21 0.07–0.50	0.010 0.007–0.023
1st year after withdrawal of ACTH	17	0.09 ± 1.12	−0.54 ± 1.12					
1st year after withdrawal of prednisolone, during ACTH treatment				10	−2.0*** ± 1.11	1.36* ± 1.40	1.36*	0.016 0.006–0.025

^aSynachten® Depot.

Levels of significance for difference from the reference population (11) are given as * $p<0.05$, ** $p<0.01$ and *** $p<0.001$.

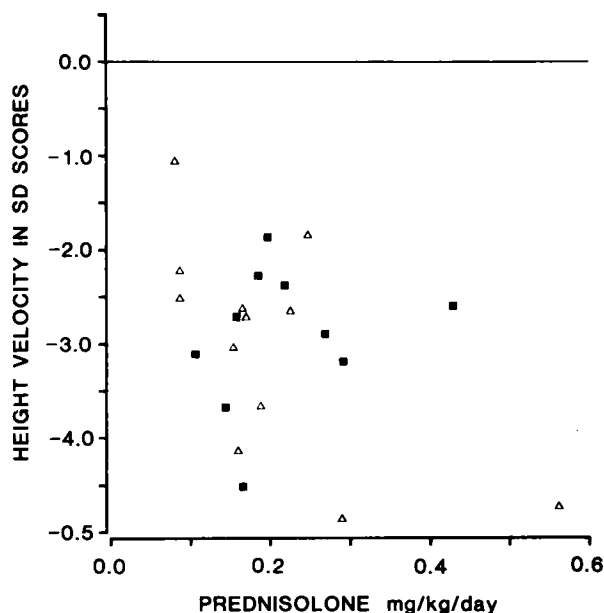


Fig. 2. Height velocity SDS during the first year of treatment in relation to the dose of prednisolone ($r = -0.39$, NS). ■ = group P+A, △ = group P.

the mean SDS for age at peak height velocity (PHV) nor the mean adult height SDS differed significantly from the reference means (12, 14) (Table 3, Fig. 3). All but one of the prednisolone-treated children were followed until adult height. In the boys the mean age at PHV was significantly higher than the reference mean (Table 3). This was not true for the

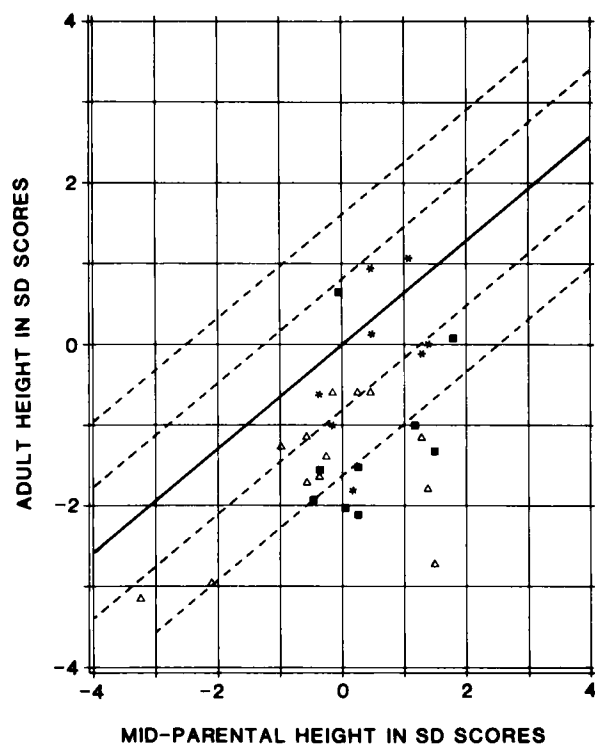


Fig. 3. Adult height SDS in relation to midparental height SDS. * = group A, ■ = group P+A, △ = group P. Mean, ± 1 SD and ± 2 SD for the reference population (publication forthcoming) are indicated by the oblique lines.

girls. Mean adult height adjusted for mid-parental height was reduced in the prednisolone-treated patients (Fig. 3). Age at menarche varied within the normal range (15) in all treatment groups, the mean values being 14.1, 13.7 and 13.5 years for groups A, P+A and P respectively.

Age at PHV did not correlate with the duration of prednisolone therapy or age at its withdrawal. Nor did adult height with these variables.

DISCUSSION

Severe asthma as such may have a negative influence on growth (7, 16, 17). This effect of the disease as such was evident in the present group of children. The acceleration of growth during the first year of ACTH treatment did not reflect pubertal growth spurt, as no child showed any sign of pubertal maturation. The dose did not correlate with height velocity. A few of the children requiring very high doses had lower growth rates, which might reflect the severity of asthma rather than the dose of ACTH. Thus the acceleration might involve catch-up growth (18) due to better control of the disease. Alternatively, it could result from growth-promoting effect of the ACTH treatment, presumably mediated by an increased secretion of adrenal androgens (19). The fact that adrenal androgens may accelerate growth is evidenced by the association of the "mid-growth spurt" (20) at about 6–8 years of age with normal adrenarche (21).

During the first year of prednisolone treatment all children improved clinically, but their height velocity was even lower than before. Despite the fairly low dosage the growth-suppressive effect was strong during the whole period of treatment, leading to a progressive decrease in height SDS. Growth was already retarded at a dose of 0.1 mg prednisolone/kg/day, which is well below previously reported limits in asthmatic children (2, 3, 4). As a consequence no dose effective in the treatment of asthma can be considered safe with regard to growth.

Long-term glucocorticoid treatment is reported to reduce the levels of testosterone in adult men (22). This might be relevant to the mechanisms behind the delayed pubertal growth spurt in the prednisolone-treated boys as the pubertal growth spurt depends on an increase in testicular androgens. However, Hauspie et al. (16) found a delay in puberty of

Table 3. *SDS of age at peak height velocity (PHV) and SDS of adult height in relation to mid-parental height for the three treatment groups*

Values are means (ranges)

	<i>n</i>	Age at peak	
		Height velocity	Adult height
Group A			
Boys	9	−0.2 (1.4–1.8)	−0.6 (1.1–1.8)
Girls	2	−0.9 (−0.7–1.1)	−0.8 (−0.6–1.0)
Group P+A			
Boys	7	−1.8* (0.2–3.0)	−1.9** (−0.6–2.1)
Girls	2	−0.4 (0.6–1.3)	−1.5 (0.1–1.6)
Group P			
Boys	3	−2.6* (−1.8–3.7)	−2.2 (−0.6–2.7)
Girls	10	−0.6 (1.6–3.2)	−1.6** (−0.6–3.1)

Levels of significance for difference from the reference population (12, 14) are given as * $p < 0.05$ and ** $p < 0.01$.

about 1.3 years in asthmatic boys not treated with glucocorticoids. Thus, the disease per se might be a causative factor in the delay of sexual maturation of the boys.

Martin et al. (17) reported normal adult height in patients with severe asthma, despite a delay in growth at around the age of 14. In the present group the pronounced growth retardation before puberty was not compensated for by the pubertal spurt. A period of increased height velocity before puberty is apparently necessary in order to reach the expected adult height. According to Prader (18) the possibility of complete or near complete catch-up growth diminishes as puberty approaches. Consequently, the age at withdrawal of glucocorticoids appears important to the final outcome of growth. In the present group no such relationship was found. However, in the four children who reached adult heights above -1 SDS, prednisolone had been withdrawn before PHV. No child on prednisolone at the time of PHV reached that level of final height. Thus, we conclude that there is probably less risk of growth stunting if glucocorticoids can be avoided during the period immediately preceding the pubertal growth spurt.

The acceleration of growth after ACTH is substituted for prednisolone is in agreement with previous findings (8, 9, 23) but was not sufficiently pronounced to compensate for the lost height potential.

It is apparent that both systemic prednisolone and ACTH interfere with the hormonal regulation of growth. ACTH might induce an enhancement of growth rate, but this seems to be transient, and adult height is not perceptibly affected. In contrast, treatment with glucocorticoids, if not terminated well in advance of puberty, entails a considerable risk of stunting.

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